

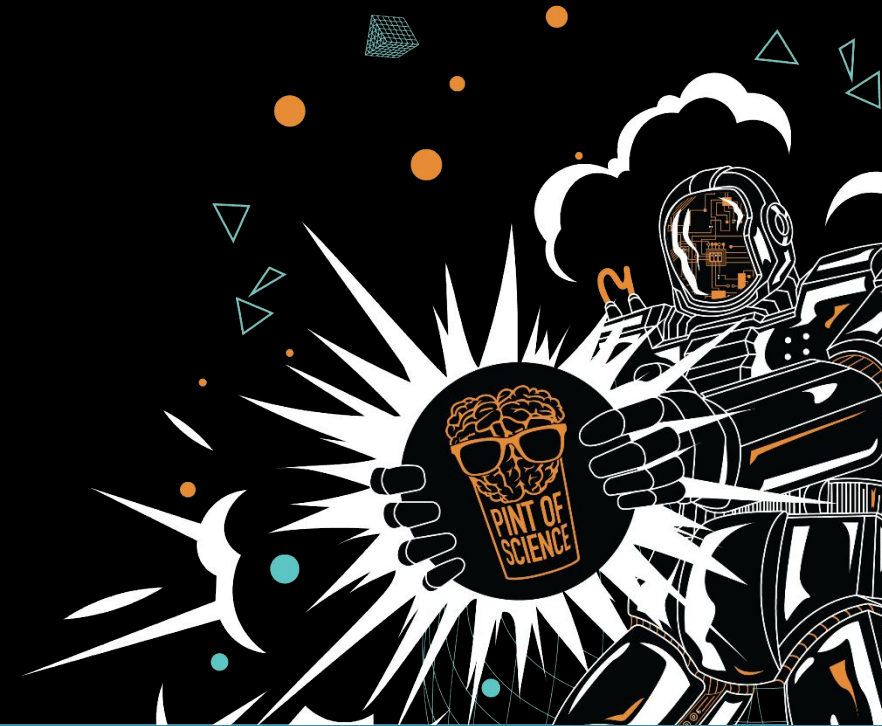


Tales of a Neotropical arachnid: Mating and Male Parenthood

Laís A. Grossel
MSc. Zoology



laisgrossel@gmail.com
laisgrossel.github.io



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Have you ever
watched this
scene?



Have you ever
watched this
scene?



Sexual Selection

Or this one?



Salome Hoogendijk



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Or this one?

Parental care



Salome Hoogendijk

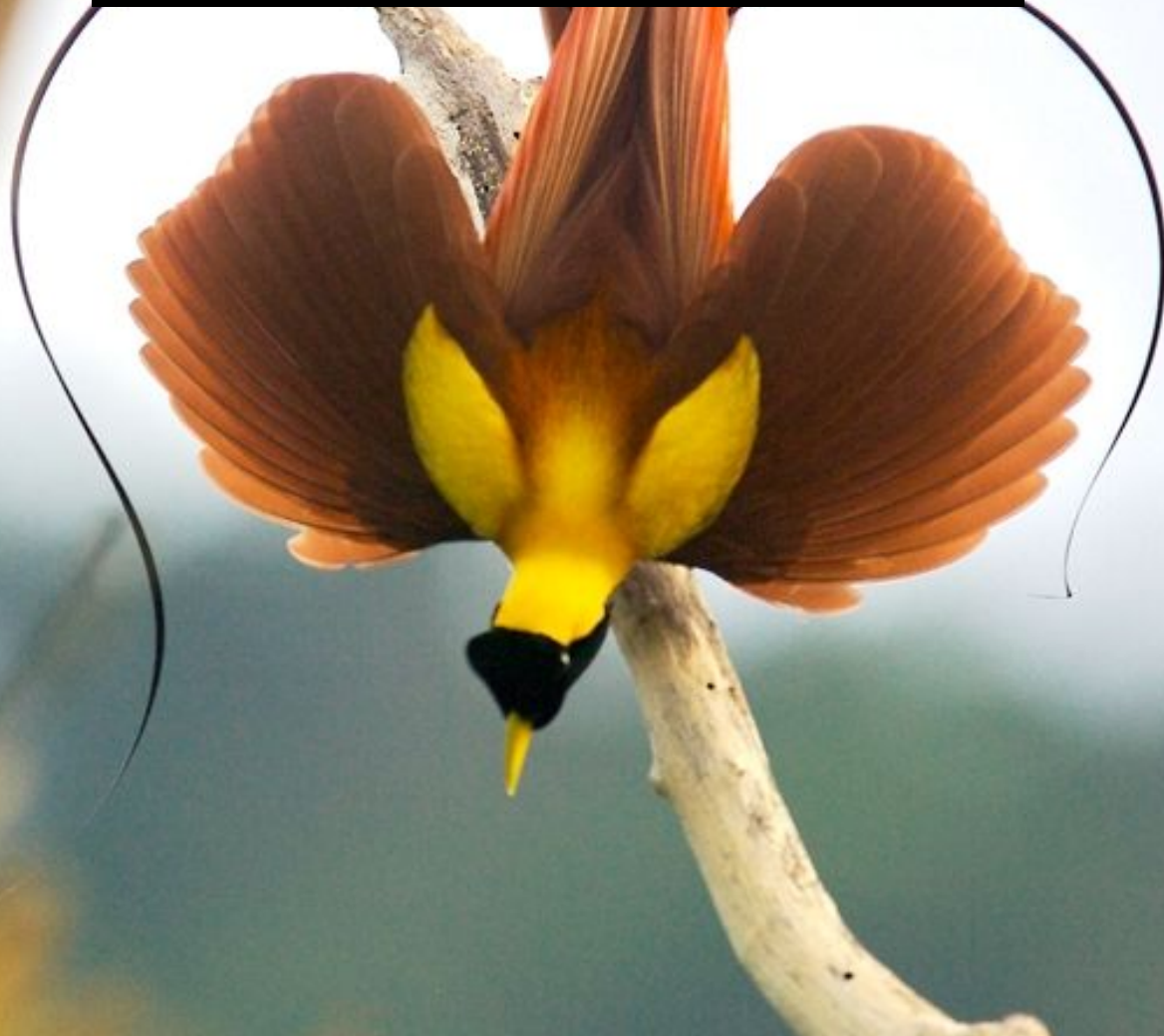


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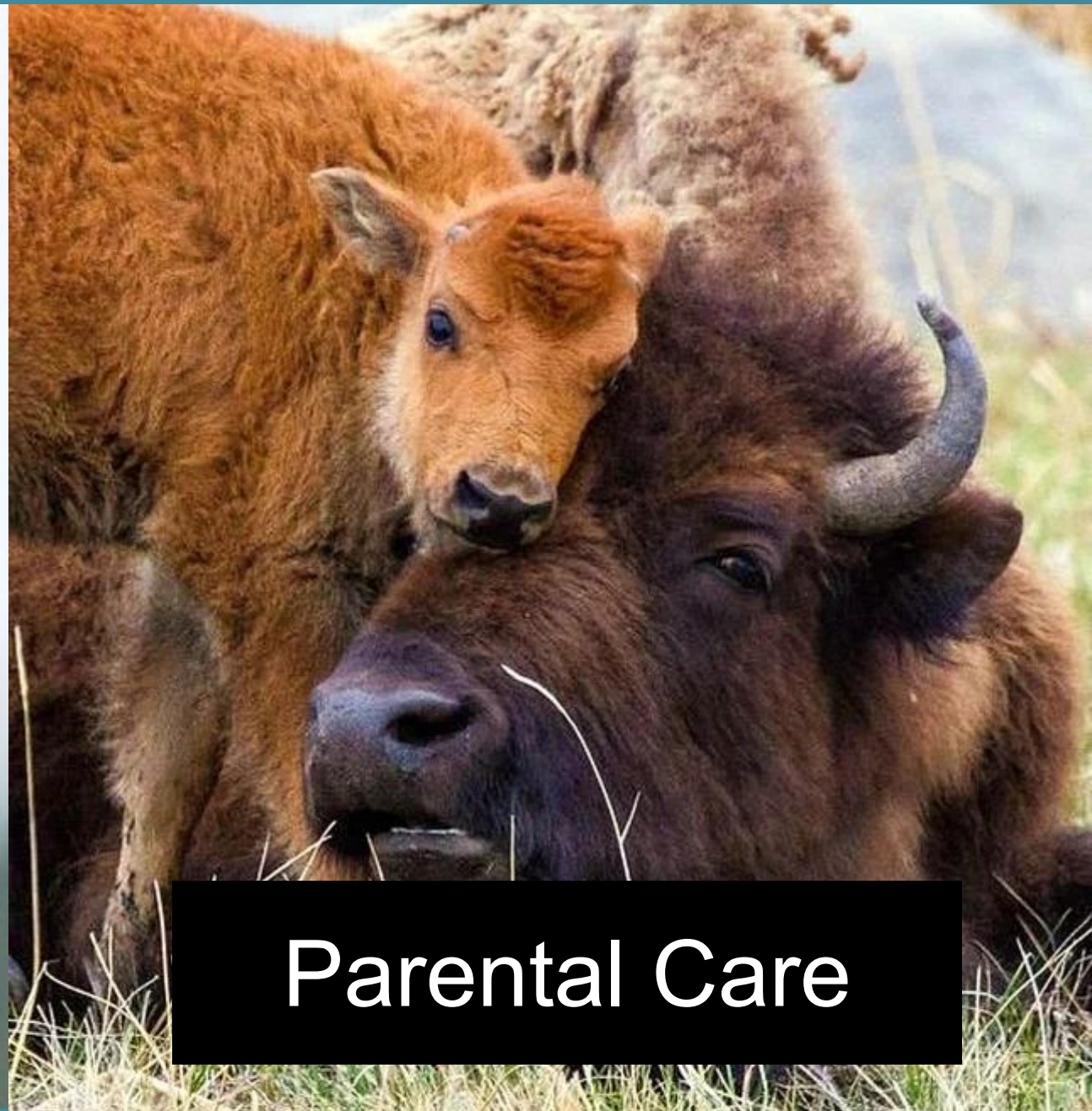
Sexual Selection



Birds of Paradise Project



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Parental Care

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Sexual Selection



Melvyn Yeo



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Josue Ramos Galdamez

Parental Care

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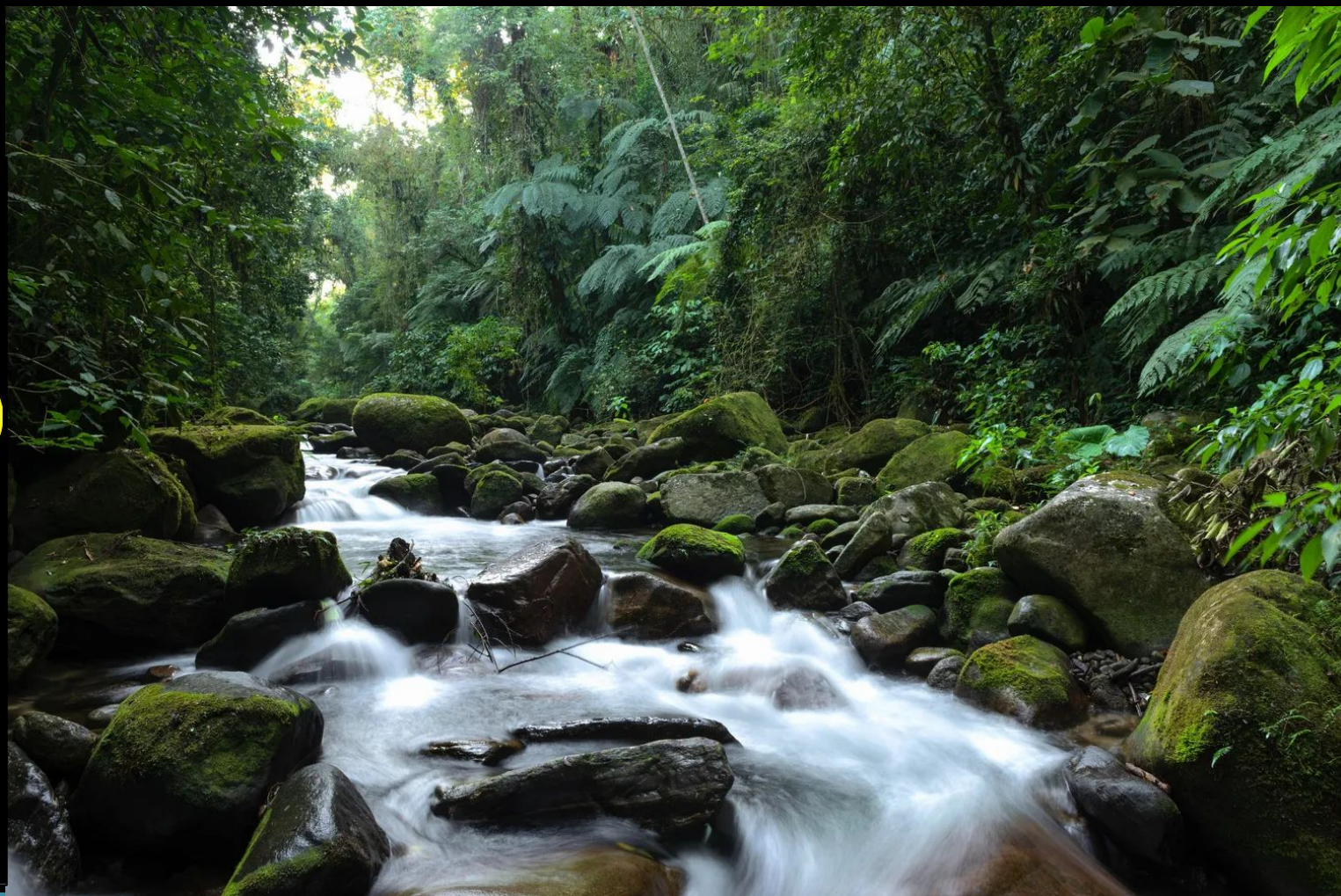
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Brazil: Atlantic Forest



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Ipomema pustulosa

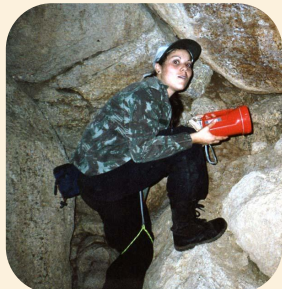


Iporangaia pustulosa

São Paulo, Brazil, 2004
You shoulda been there!

Once Upon a Time a Harvestman

by Quentin Tarantino

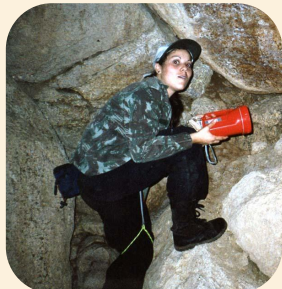


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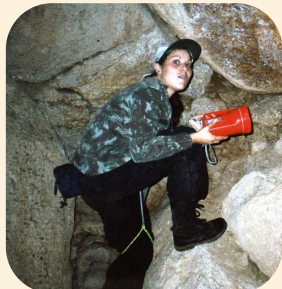


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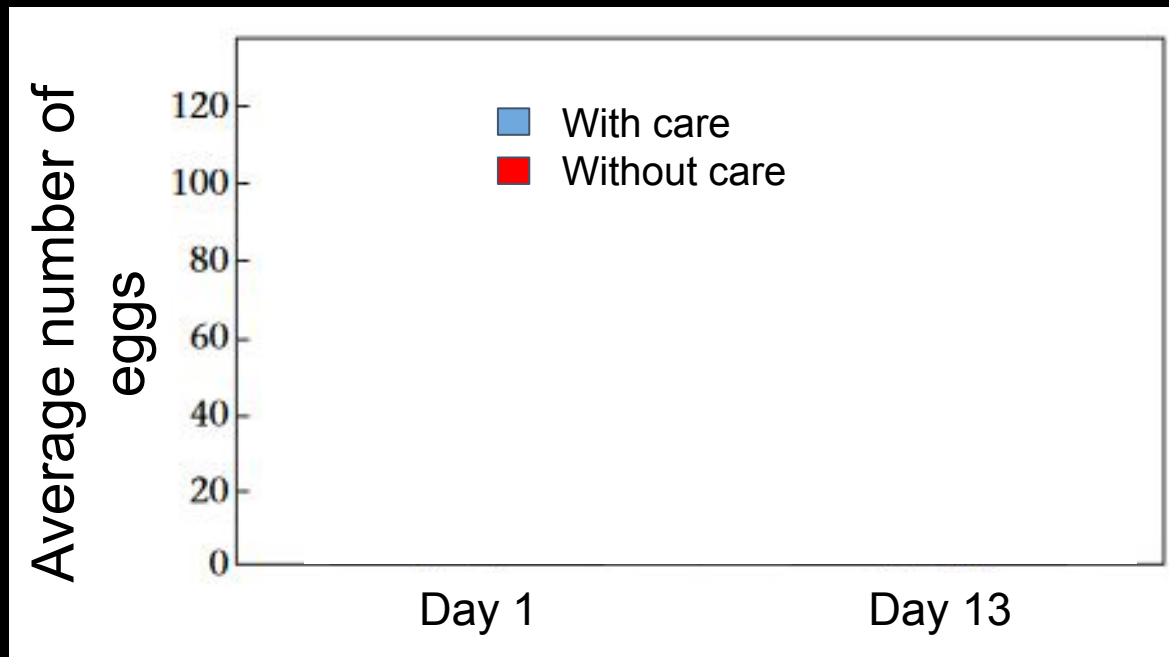
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Is the care efficient?

Requena et al. 2009, *Animal Behaviour*



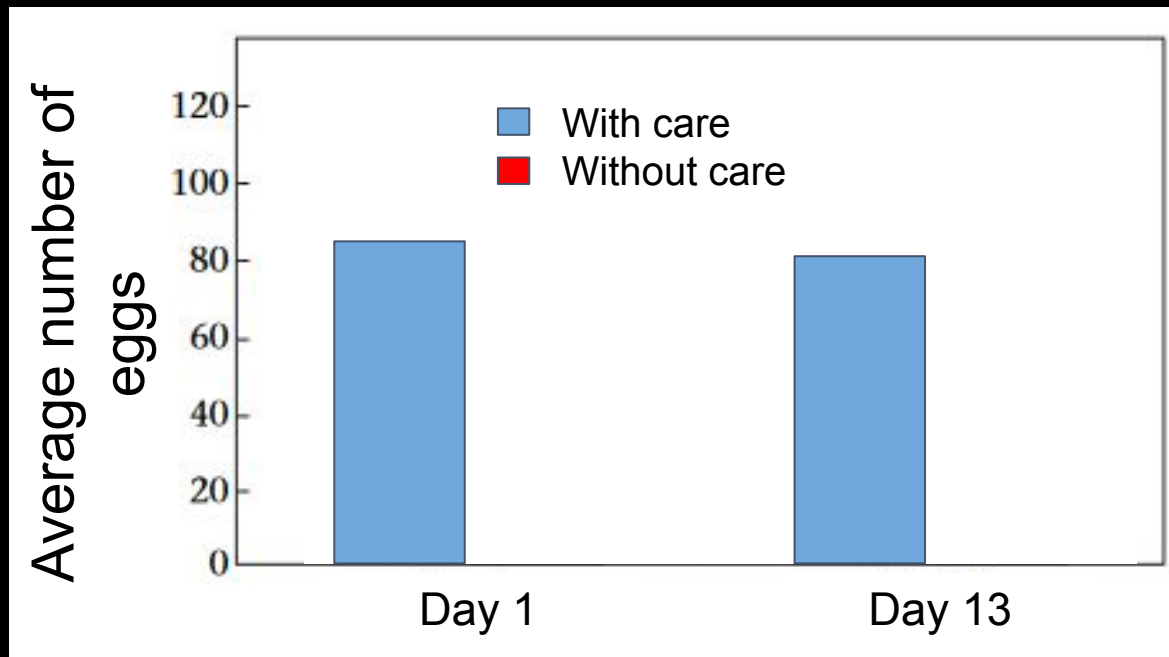
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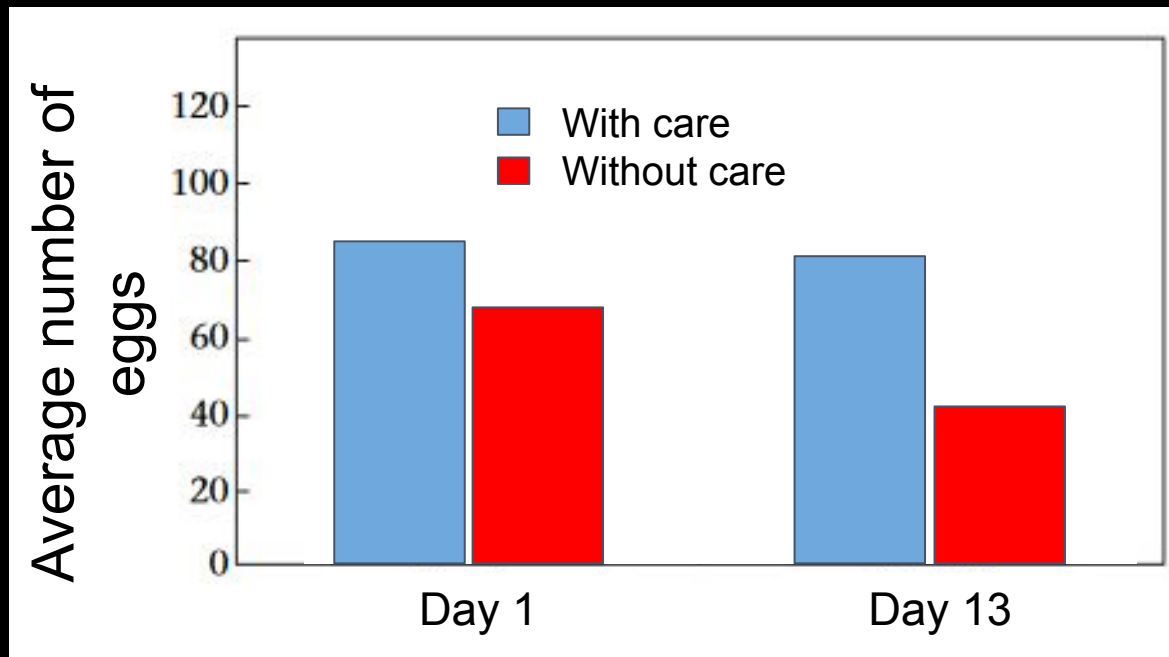
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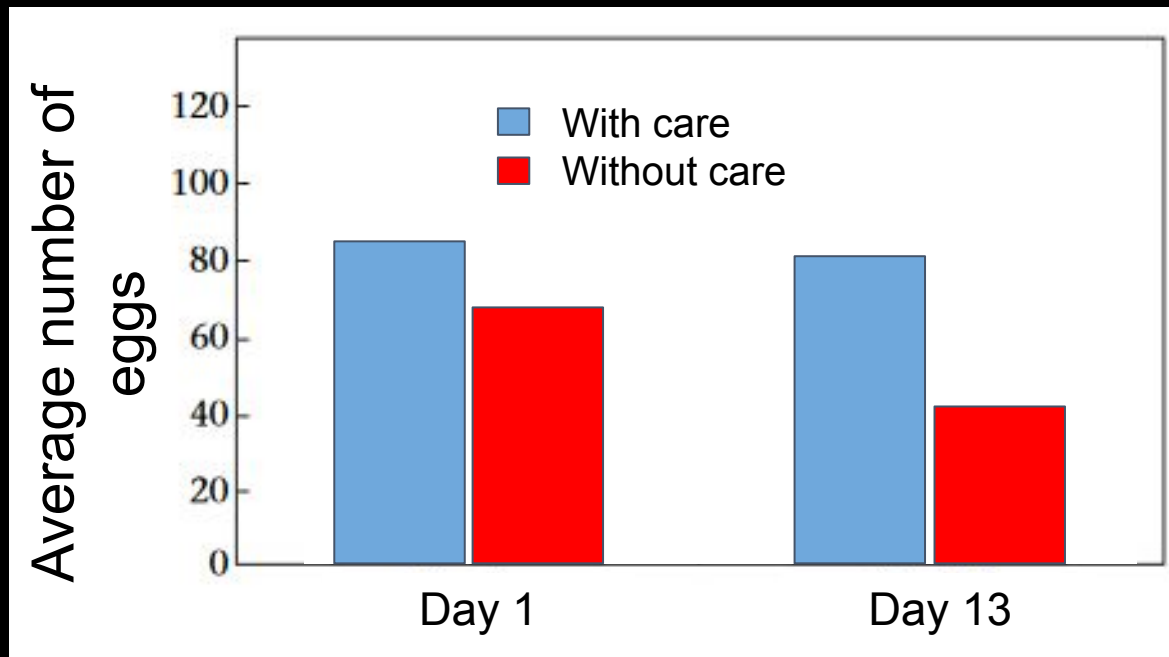
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Adjustment of paternal care

Predation



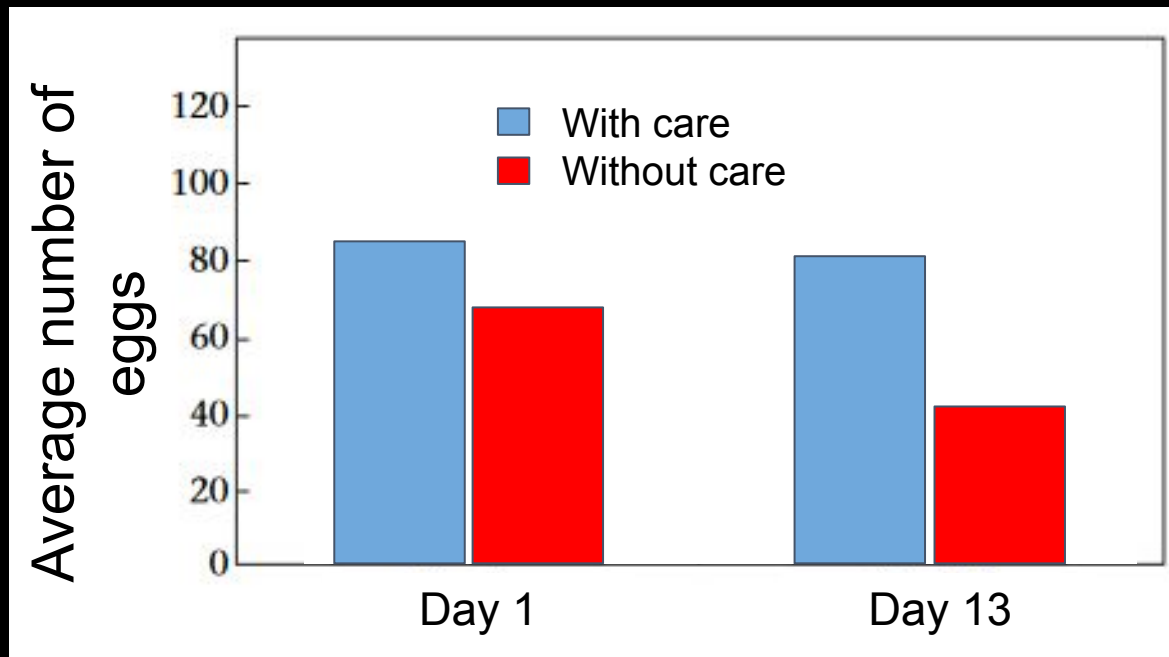
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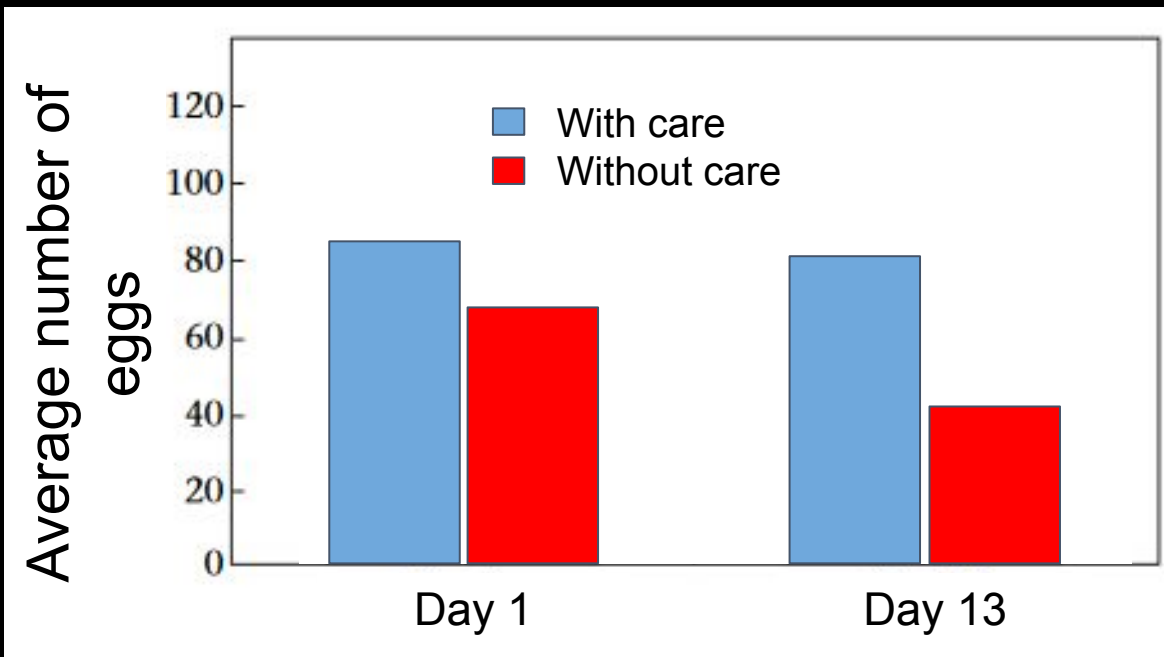
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Adjustment of paternal care



Predation

Continuous
care



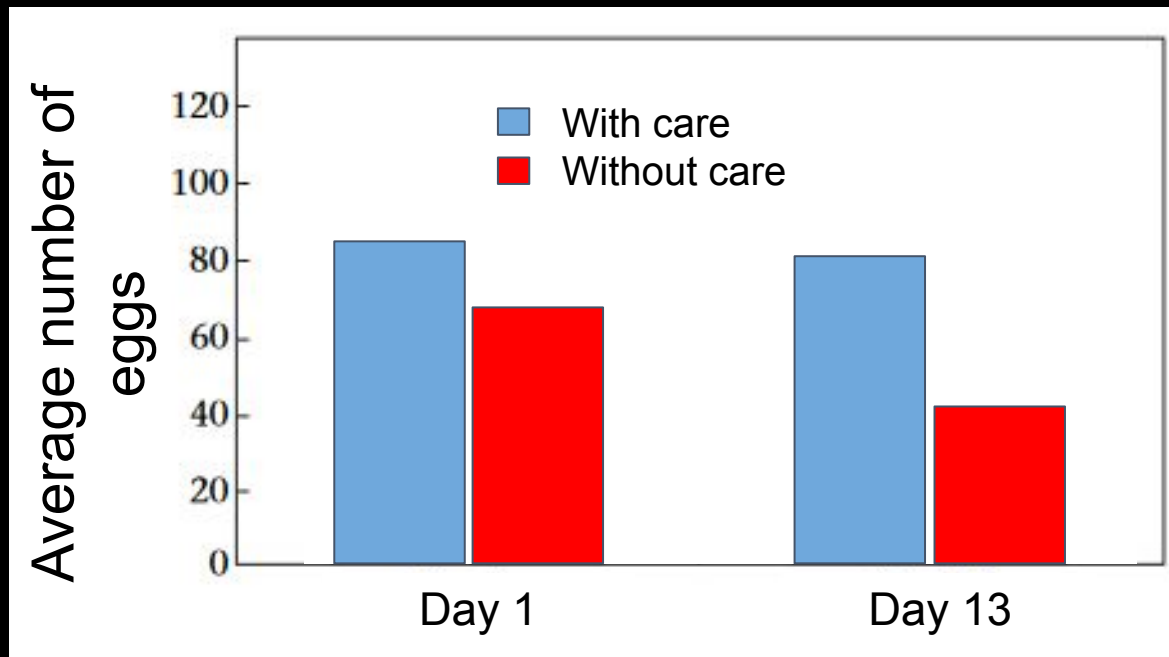
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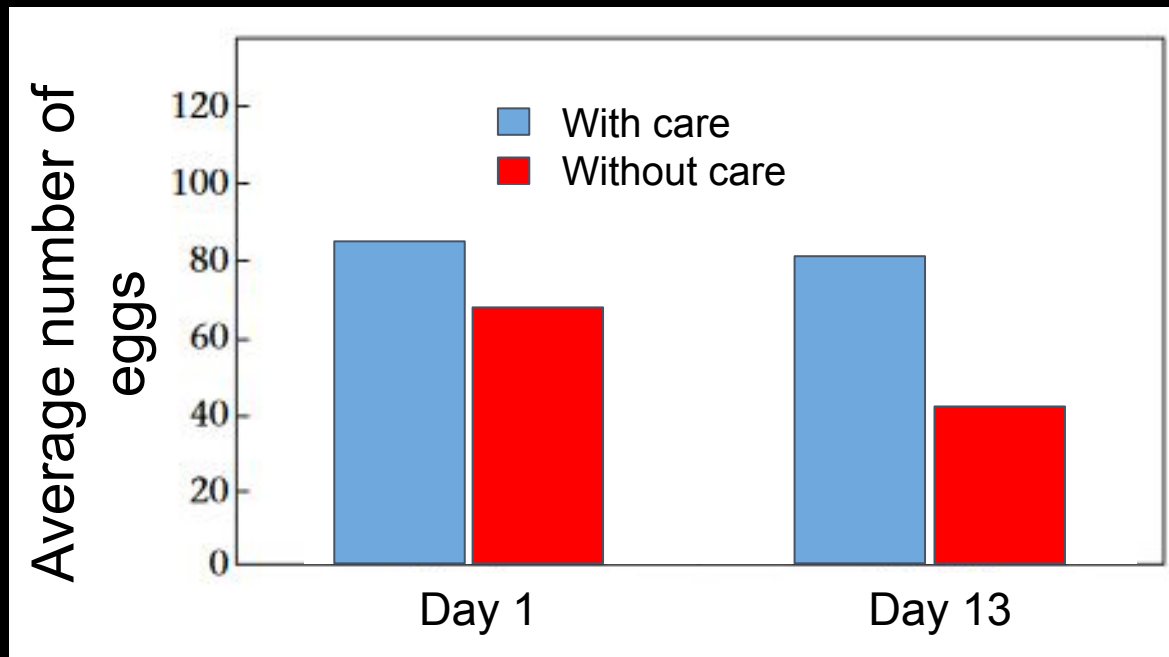
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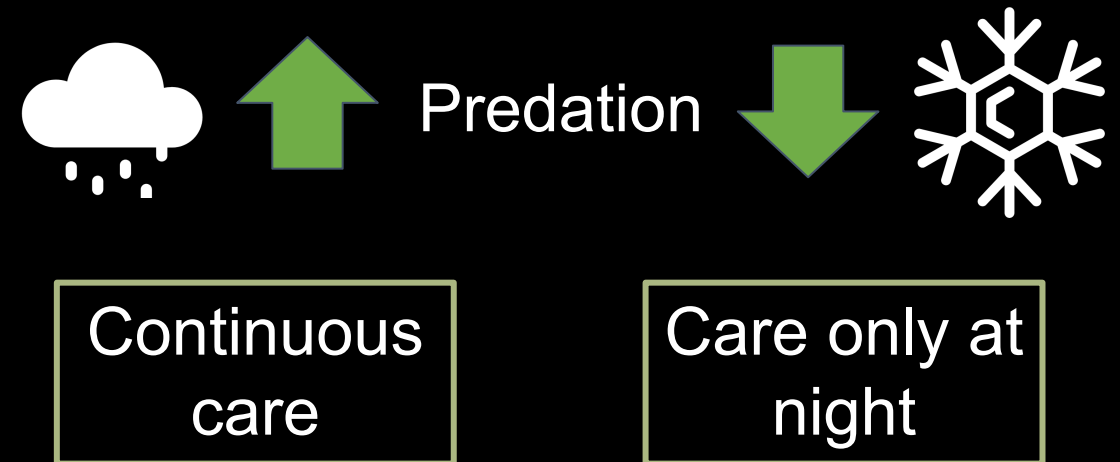
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Adjustment of paternal care



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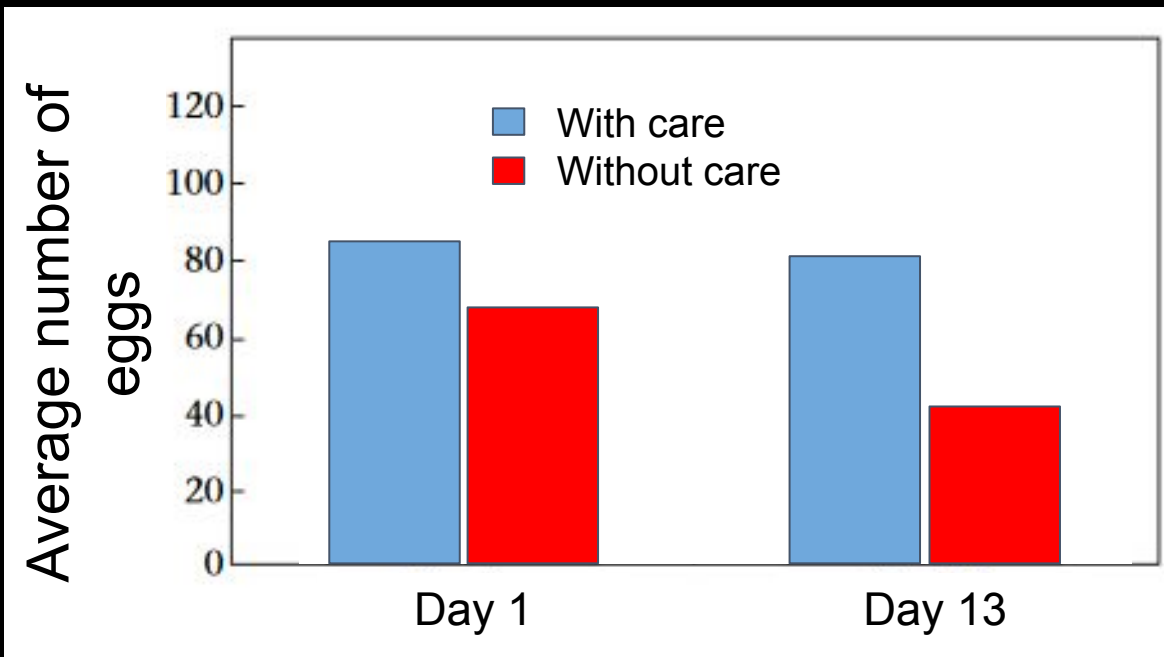
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Is the care efficient?

Requena et al. 2009, *Animal Behaviour*

It increases offspring survival



Adjustment of paternal care



Continuous care

Care only at night



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Is the care costly?

Requena et al. 2012, *PLOS One*

Proportion of feeding individuals

Probability of surviving



Females



Non-caring males



Caring males



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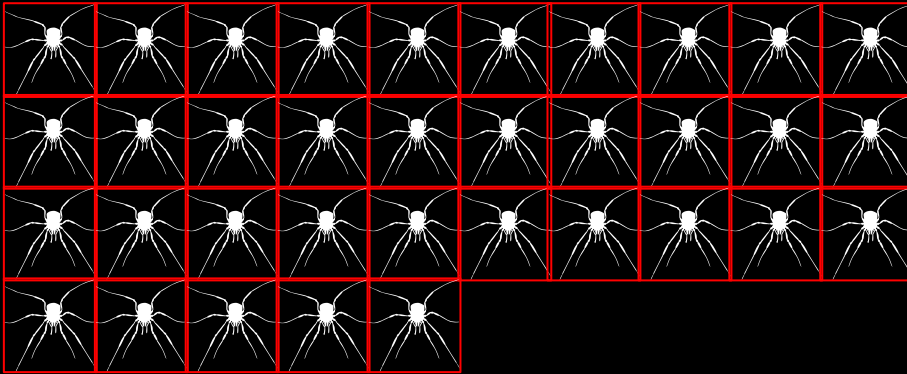
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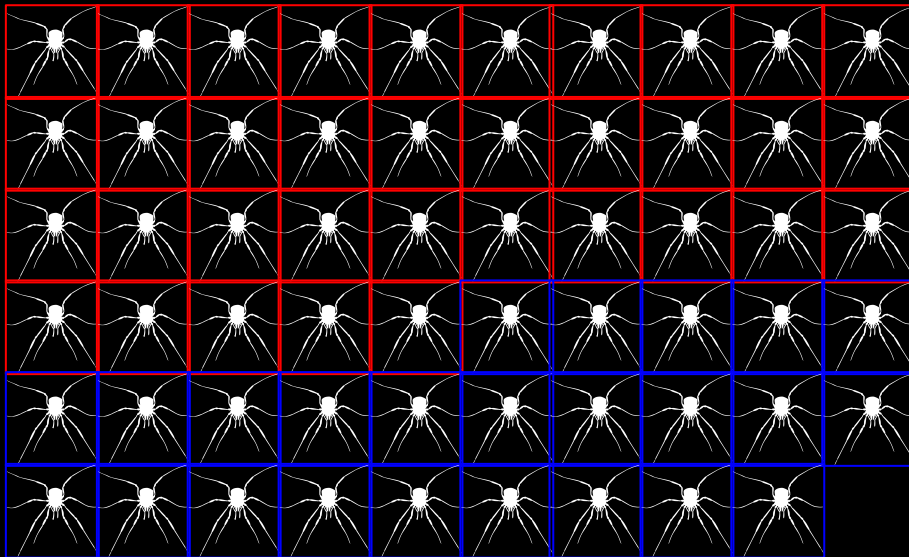
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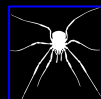
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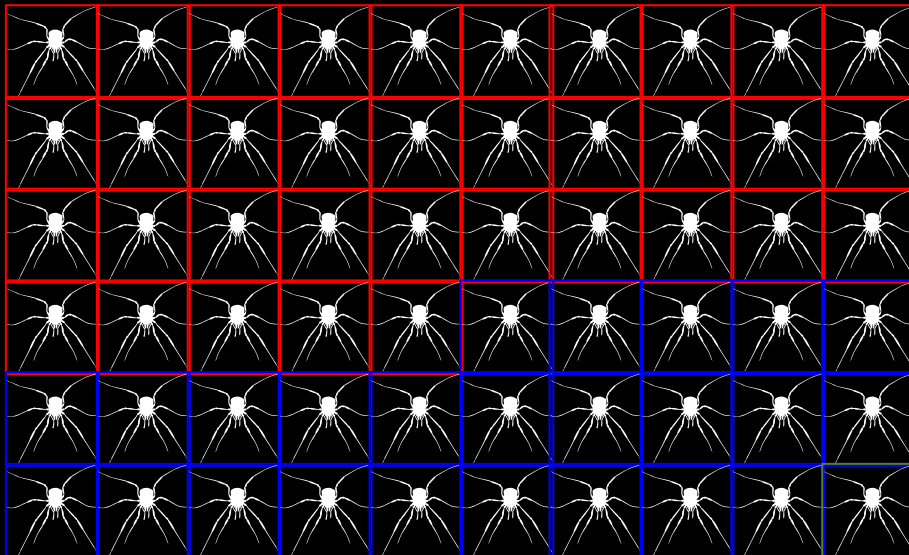
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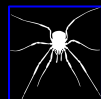
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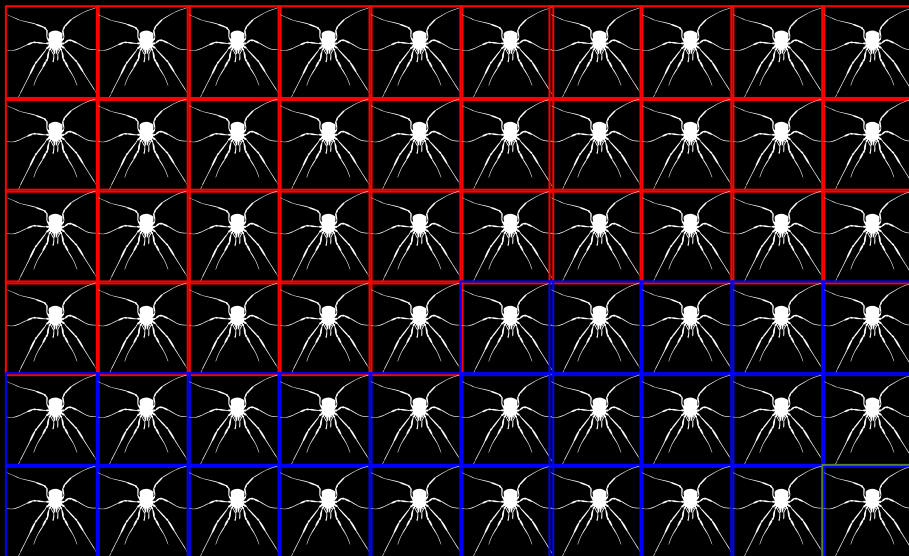
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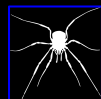
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Probability of surviving



Opilio Tracker



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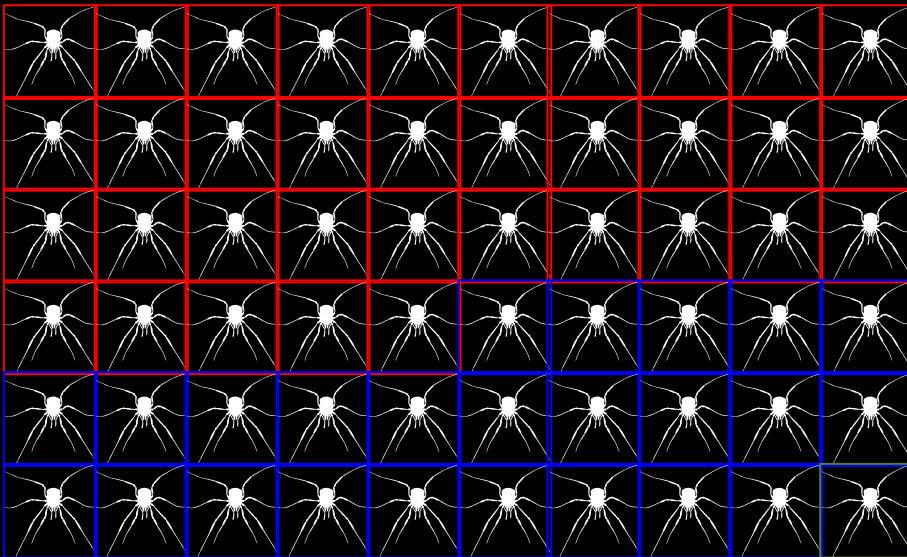
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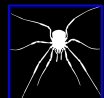
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Caring males

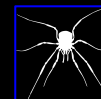
Probability of surviving



Opilio Tracker



Females



Non-caring males



Caring males



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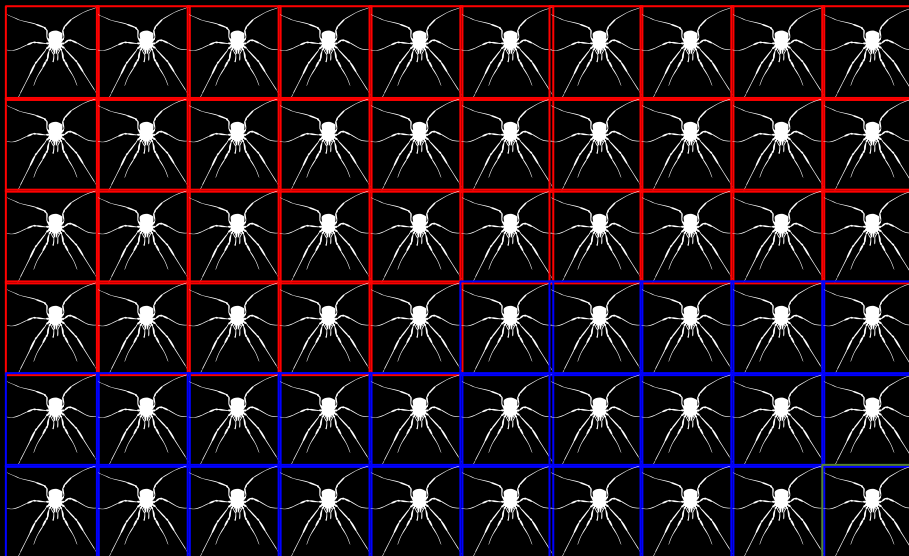
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Requena et al. 2012, *PLOS One*

Proportion of feeding individuals



Females



Non-caring males

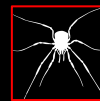


Caring males

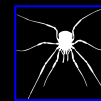
Probability of surviving



Opilio Tracker



Females



Non-caring males



Caring males



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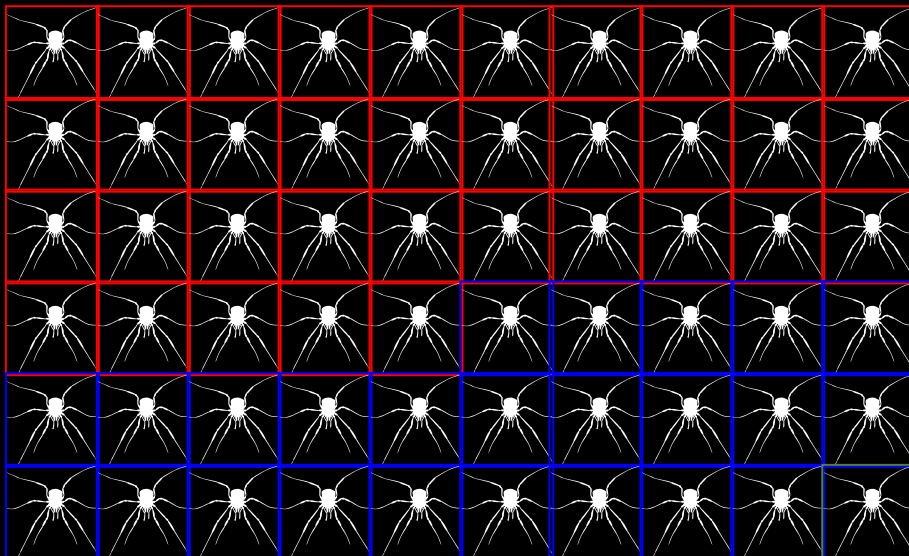
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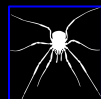
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Proportion of feeding individuals



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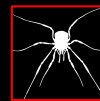
Caring males

In terms of body condition, but not survival

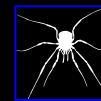
Probability of surviving



Opilio Tracker



Females



Non-caring males



Caring males



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How do females choose males?

Requena & Machado 2015, *Behavioural Ecology*



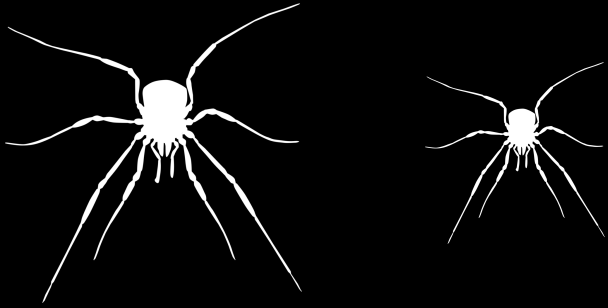
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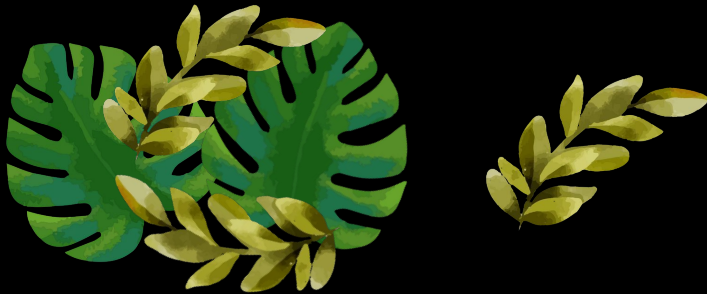
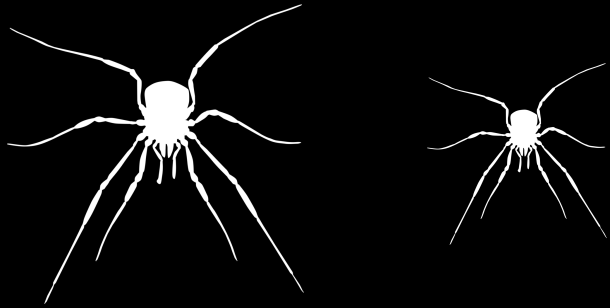
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How do females choose males?

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Males with eggs

Males without eggs



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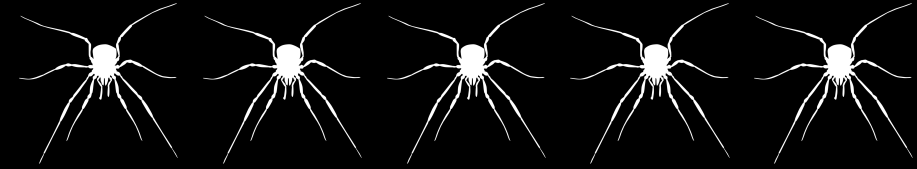
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Requena & Machado 2015, *Behavioural Ecology*



Males with eggs



Males without eggs



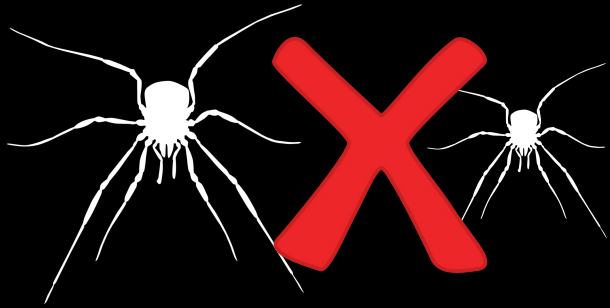
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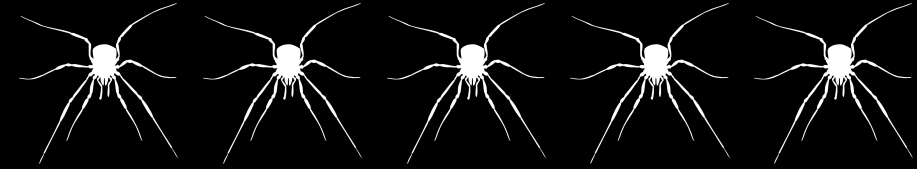
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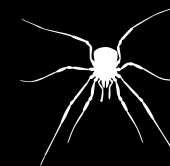
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Males with eggs



Males without eggs



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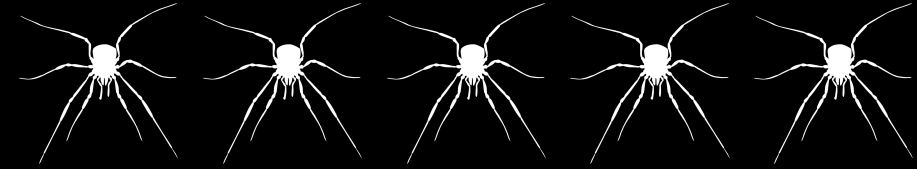
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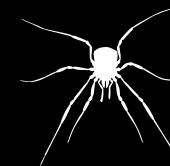


Females prefer
the daddies

Males with eggs



Males without eggs



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Is there variation in the quality of care?

Alissa et al. 2024, *Behavioral Ecology and Sociobiology*



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Males in GOOD condition

Males in POOR condition



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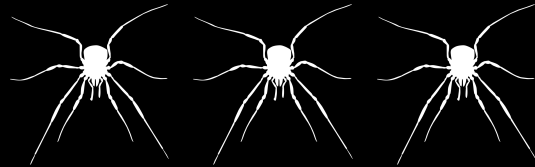
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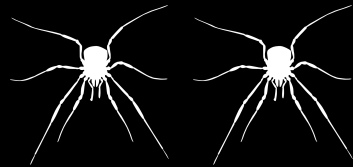
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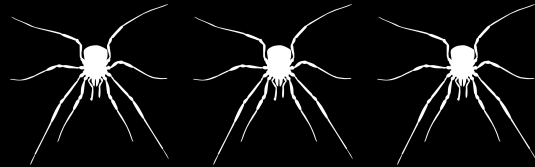
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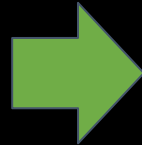
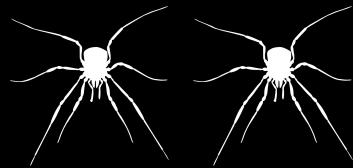
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Proportion of clutch lost



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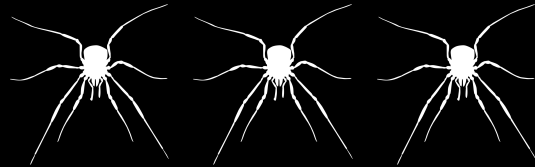
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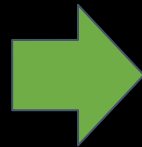
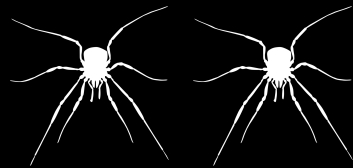
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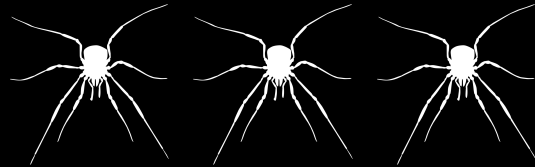
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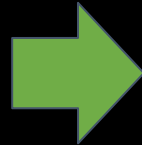
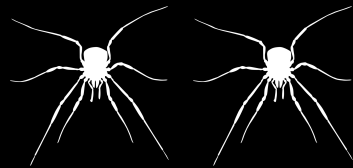
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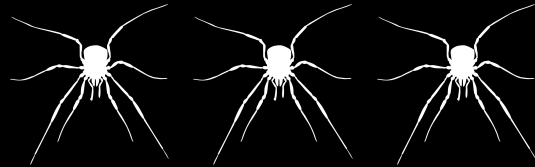
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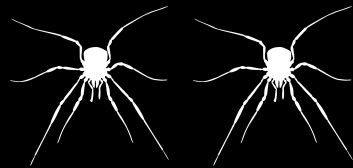
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Proportion of clutch lost

Fit dads are sexier...



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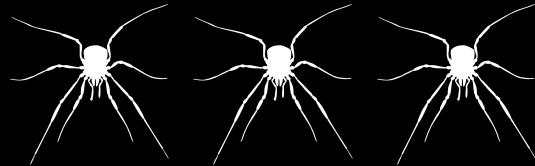
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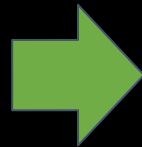
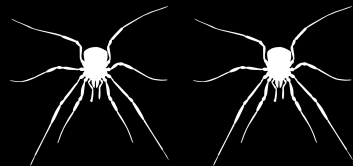
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Males in POOR condition



Proportion of clutch lost

Fit dads are sexier...

...but they're not
better caregivers!



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Then why are males in good condition chosen?

Grossel & Machado 2025, *Behavioral Ecology and Sociobiology*



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Grossel & Machado 2025, *Behavioral Ecology and Sociobiology*

GOOD condition



POOR condition



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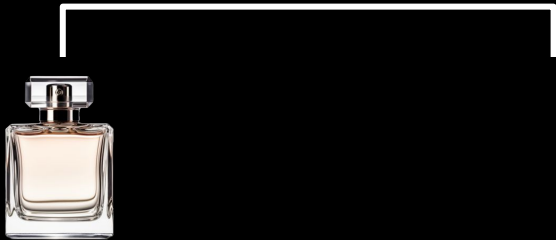
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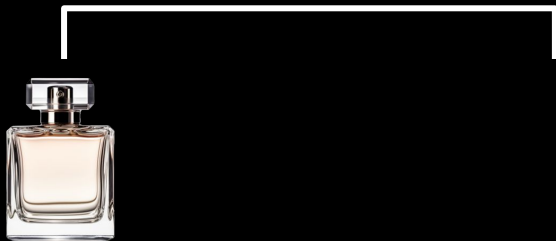
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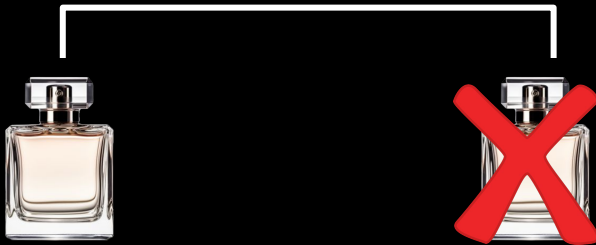
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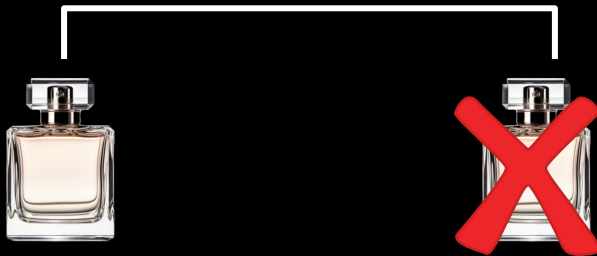
Then why are males in good condition chosen?

Grossel & Machado 2025, *Behavioral Ecology and Sociobiology*

GOOD condition



POOR condition



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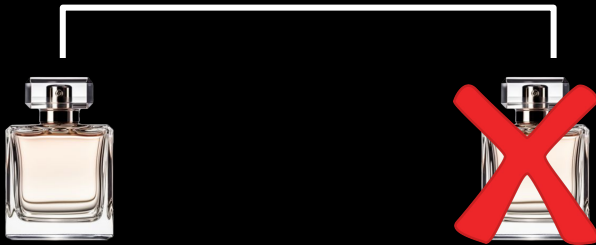
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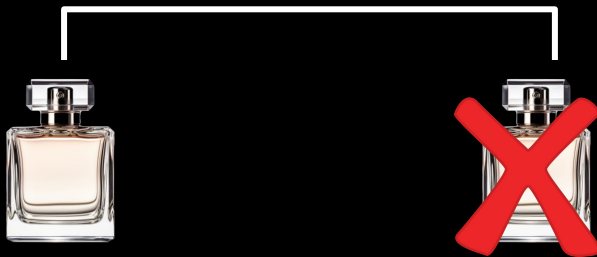
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GOOD



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Then why are males in good condition chosen?

Grossel & Machado 2025, *Behavioral Ecology and Sociobiology*



GOOD



1

2

3

POOR



4



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Then why are males in good condition chosen?

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Then why are males in good condition chosen?

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Because they advertise better their better condition



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Take home messages

2004

2009

2012

2015

2024

2025

Exclusive
paternal care



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Take home messages

Care increases survival
(**of clutches and
parental males**)

2004

2009

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Exclusive
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Natural Selection

Care increases survival
(**of clutches and
parental males**)

2004

2009

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Natural Selection

Care increases survival
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Exclusive
paternal care



Females prefer **parental males**,
in **good condition**
and able **to advertise**



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Natural Selection

Care increases survival
(**of clutches and
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Sexual Selection

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Females prefer **parental males**,
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Andy Harmsen



Steve Kerr



Ryosuke Kuwahara



Jen Cross

© Jen Cross



Damien Broule



Tiago G. C. Antunes



Ken-ichi Ueda



Tony Iwane



Gonzalo Giribet



George Mitchell



gernotkunz



Ken-ichi Ueda

Thank you!



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Art by Diego Solano-Brenes



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